CSI 2132 – Databases A

- Normalization and Functional Dependencies – Additional Exercises

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Exercise 1

Consider the relation schema R = (A, B, C, D, E) and FDs the set F:

$$F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$$

Compute the canonical cover and the minimal cover of F.

Exercise 1 - solution

For the computation of the canonical cover:

- We try to union FDs with the same LHS
- We try to remove extraneous attributes
- We do the above steps until there is no change in the set of FDs that we have

We start with the original FDs: F = {A → BC, CD → E, B → D, E → A}

We cannot perform union.

We check for extraneous attributes. Let's take A → BC and check whether B or C are extraneous. If B is extraneous, then calculating A+ with respect to F' = {A → C, CD → E, B → D, E → A} should include B. But A+ = AC. If C is extraneous, then calculating A+ with respect to F' = {A → B, CD → E, B → D, E → A} should include B. But A+ = ABD. Therefore A → BC does not have extraneous attributes.

Exercise 1 - solution

For the computation of the canonical cover:

- We try to union FDs with the same LHS
- We try to remove extraneous attributes
- We do the above steps until there is no change in the set of FDs that we have

We continue the check for extraneous attributes. Let's take CD → E and check whether C or D are extraneous. We calculate C+ and D+ based on F: C+ = C and D+ = D. Therefore CD → E does not have any extraneous attributes.

Thus the canonical cover is Fc = {A → BC, CD → E, B → D, E → A}

And the minimal cover is Fm = = {A → B, A → C, CD → E, B → D, E → A}

Exercise 2

Consider the relation schema R = (A, B, C, D, E) and FDs the set F:

$$F = \{A \rightarrow BC, B \rightarrow C, AB \rightarrow D\}$$

Compute the canonical cover and the minimal cover of F.

Exercise 2 - solution

- We first try to union FDs with common LHS. This is not possible in the original $F = \{A \rightarrow BC, B \rightarrow C, AB \rightarrow D\}$
- We look for extraneous attributes. Let's do the check for $AB \rightarrow D$:
  - First we check if A is extraneous. We find the closure of B, B+ with respect to F: $B^+ = BC$. Therefore, B+ does not include D, so A is not extraneous.
  - We check if B is extraneous. We need to find the closure A, A+ with respect to F: $A^+ = ABCD$. This includes D, so B is extraneous.
- We eliminate the extraneous attribute and the set of FDs is now:

$$F1 = \{A \rightarrow BC, B \rightarrow C, A \rightarrow D\}$$

Exercise 2 - solution

$$F1 = \{A \rightarrow BC, B \rightarrow C, A \rightarrow D\}$$

- We perform the union and the set becomes:

$$F2 = \{A \rightarrow BCD, B \rightarrow C\}$$

- We continue the check of extraneous attributes for $A \rightarrow BCD$.
  - We check if B is extraneous. We calculate A+ based on F1' $\{A \rightarrow CD, B \rightarrow C\}$. A+ = ACD, therefore B is not extraneous.
  - We check if C is extraneous. We calculate A+ based on F1' $\{A \rightarrow BD, B \rightarrow C\}$. A+ = ABCD, therefore C is extraneous.
  - We check if D is extraneous. We calculate A+ based on F1' $\{A \rightarrow BC, B \rightarrow C\}$. A+ = ABC, therefore D is not extraneous.

- The set of FDs becomes $F3 = \{A \rightarrow BD, B \rightarrow C\}$. There are no more changes to be done. Therefore $Fc = F3 = \{A \rightarrow BD, B \rightarrow C\}$ and $Fm = \{A \rightarrow B, A \rightarrow D\ B \rightarrow C\}$.

Exercise 3

Consider a relation schema R (A, B, C, D) and FDs the set F:

$$F = \{C \rightarrow D, C \rightarrow A, B \rightarrow C\}$$

a. Identify the candidate keys of R.
b. What is the highest normal form that R satisfies?
c. If R is not in 3NF, decompose it into 3NF.
d. If R is not in BCNF, decompose it into BCNF.

Exercise 3 - solution

$R = (A, B, C, D)$.

$F = \{C \rightarrow D, C \rightarrow A, B \rightarrow C\}$.

We find set $\mathcal{A}$ that contains all attributes that do not appear on the RHS of any FD: $\mathcal{A} = \{B\}$

We find set $\mathcal{B}$ that contains all attributes that appear on the RHS of some FD, but not on the LHS: $\mathcal{B} = \{A, D\}$.

We find the closure of $\mathcal{A}$:

$$
\begin{aligned}
B^+ &= B && (B \rightarrow B) \\
&= BC && (B \rightarrow C) \\
&= BCD && (C \rightarrow D) \\
&= ABCD && (C \rightarrow A)
\end{aligned}
$$

Since B^+ contains all the attributes of the relation schema, B it is the only candidate key

Exercise 3 - solution

$R = (A, B, C, D)$.

$F = \{C \rightarrow D, C \rightarrow A, B \rightarrow C\}$.

R is not 3NF, because:

- $C \rightarrow D$ causes a violation: it is non-trivial, C is not a superkey, D is not part of any candidate key.
- $C \rightarrow A$ causes a violation: is non-trivial, C is not a superkey, A is not part of any candidate key.
- $B \rightarrow C$ does not cause violation, because B is a superkey.

R is not in 3NF and, therefore it is not in BCNF

Exercise 3 - solution

$R = (A, B, C, D)$.

$F = \{C \rightarrow D, C \rightarrow A, B \rightarrow C\}$.

In order to decompose R in 3NF we need to find the canonical cover:
Using the union operation $C \rightarrow D$ and $C \rightarrow A$ become $C \rightarrow DA$, and F becomes $F = \{C \rightarrow DA, B \rightarrow C\}$.
We check for extraneous attributes in $C \rightarrow DA$, i.e. we check whether D or A are extraneous.

- If we remove D and follow the respective algorithm which says: " Find $\alpha+$ using the dependencies in F' where $F' = (F - \{\alpha \rightarrow \beta\}) \cup \{ \alpha \rightarrow (\beta - A) \}$." Here $\alpha = C$ and $F' = \{C \rightarrow A, B \rightarrow C\}$.
- We find that $C+ = AC$, i.e. does not contain D, therefore D is not extraneous.
- We remove A and follow the algorithm. Here $\alpha = C$ and $F' = \{C \rightarrow D, B \rightarrow C\}$.
- We find that $C+ = CD$, i.e. does not contain A, therefore A is not extraneous.

Therefore $Fc = \{C \rightarrow DA, B \rightarrow C\}$.

Exercise 3 - solution

$R = (A, B, C, D)$.

$F = \{C \rightarrow D, C \rightarrow A, B \rightarrow C\}$.

$Fc = \{C \rightarrow DA, B \rightarrow C\}$.

For each functional dependency in Fc we create a new relation schema:

R1 = (C, D, A), R2 = (B, C).

The schema R2 contains the candidate key for R, therefore there is nothing more to be done.

Exercise 3 - solution

$R = (A, B, C, D)$.

$F = \{C \rightarrow D, C \rightarrow A, B \rightarrow C\}$.

We decompose R into a set of BCNF relations

$C \rightarrow D$ and $C \rightarrow A$ both cause BCNF violations.

Based on $C \rightarrow D$ we decompose R into R1= (A, B, C) and R2=(C, D).

We check for violations in R1 and R2. To do this we need to calculate the closure of F, F+. We follow one of the two respective algorithms for this.

$F+ = \{C \rightarrow D, C \rightarrow A, C \rightarrow DA, B \rightarrow C, B \rightarrow D, B \rightarrow A, B \rightarrow CD, B \rightarrow DA, B \rightarrow CA, B \rightarrow CDA, AB \rightarrow C, AB \rightarrow D, AB \rightarrow CD, AC \rightarrow D, BC \rightarrow A, BC \rightarrow D, BC \rightarrow AD, BD \rightarrow C, BD \rightarrow A, BD \rightarrow AC, CD \rightarrow A, BCD \rightarrow A, ABD \rightarrow C, ABC \rightarrow D\}$.

$F1+ = \{C \rightarrow A, B \rightarrow C, B \rightarrow A, B \rightarrow CA$ etc$\}$. Based on F1+, the candidate key of R1 is B.

R1 still violates BCNF because of $C \rightarrow A$ (already in F).

We decompose R1 to R11 = {B, C} R12 = {C, A}.

Final decomposition: R2 = {C, D}, R11 = {B, C}, R12 = {C, A}.

The decomposition is dependency preserving.

Exercise 4

$R = (A, B, C, D)$

$F = \{AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B\}$

a. Find the candidate keys of R.
b. Is R in 3NF? If not, decompose it into 3NF.
c. Is R in BCNF? If not, decompose it into BCNF.

Exercise 4 - solution

$R = (A, B, C, D)$

$F = \{AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B\}$

- We find the candidate keys. All attributes, A,B,C,D appear in both the LHS and the RHS of FDs. Therefore, there are no attributes that are certainly part of all candidate keys or are certainly not part of any candidate key.
- We start with singleton attributes and calculate their closures. If we don't find a key, we continue by adding one more attribute every time to each attribute:
- We first calculate A+ = A, B+= B, C+ = C and D+ = D.
- We continue with pairs of attributes:

  - AB+ = ABCD – therefore AB is a candidate key.

  - AC+ = AC – therefore AC is not a candidate key.

  - AD+ = ABCD – therefore AD is a candidate key.

Exercise 4 - solution

$R = (A, B, C, D)$

$F = \{AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B\}$

- We do not check ACD+ or ACB+, because we have already found that AD and AB are candidate keys, therefore ACD and ACB are superkeys.
- We continue with pairs of B that we have not yet checked:
  - BC+ = ABCD – therefore BC is a candidate key.
  - BD+ = BD – therefore BD is not a candidate key.
- We do not check BCD+ or BCA+, because we have already found that BC and AB are candidate keys, therefore BCD and BCA are superkeys.
- We continue with pairs of C that we have not yet checked:
  - CD+ = ABCD – therefore BC is a candidate key.

Exercise 4 - solution

$$R = (A, B, C, D)$$

$$F = \{AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B\}$$

- Therefore the candidate keys of R are: AB, BC, CD, AD.

All FDs in F satisfy the definition of 3NF:

- in $AB \rightarrow C$, the LHS is a key
- in $AB \rightarrow D$, the LHS is a key
- in $C \rightarrow A$, the RHS is a prime attribute
- in $D \rightarrow B$, the RHS is a prime attribute

Therefore R is in 3NF.

Exercise 4 - solution

$$R = (A, B, C, D)$$

$$F = \{AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B\}$$

- Therefore the candidate keys of R are: AB, BC, CD, AD.

Two FDs in F satisfy the definition of BCNF and two don't:

- in $AB \rightarrow C$, the LHS is a key
- in $AB \rightarrow D$, the LHS is a key
- in $C \rightarrow A$, the LHS is not a key
- in $D \rightarrow B$, the LHS is not a key

Therefore R is not in BCNF.

Exercise 4 - solution

$R = (A, B, C, D)$

$F = \{AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B\}$

We decompose R in BCNF:

- We choose $C \rightarrow A$ in order to decompose into:

  R1 = (C,A) and R2 = (C, B, D).

  R1 is in BCNF with key C.
- We need to find the candidate keys of R2 in order to see if it is in BCNF. C+ = C, B+ = B, and D+ = BD (no key from the closure of singleton attributes). We check the closure of pairs of attributes: CB+ = CB, CD+ = CDB and BD+ = BD. Therefore the only key of R2 is CD.
- We need to find F+ and the subset of F+, that holds in R2 i.e. F2+.

Exercise 4 - solution

$R = (A, B, C, D)$

$F = \{AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B\}$

Decomposition until this point: R1 = (C,A) and R2 = (C, B, D)

- F+ = {AB → C, AB → D, AB → CD, AD → B, AD → C, AD → BC, BC→A, BC→D, BC→AD, CD→A, CD→B, CD→AB, ABC→D, ACD→B, BCD→A, C → A, D → B}, and the subset of it that holds in R2 is F2+ = {D → B etc} (Note that we could find that the candidate key of R2 is CD by observing F2 and concluding that D and C are certainly part of any candidate key and B is not).
- Therefore R2 is not in BCNF because in D → B the LHS is not a key.
- We decompose R2 to: R21 = (D, B) and R22 = (D, C). The key of R21 is D, the key of R22 is DC and both are in BCNF,
- The final decomposition is: R1 = (C,A), R21 = (D, B) and R22 = (D, C).

Exercise 4 - solution

$R = (A, B, C, D)$

$F = \{AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B\}$

Final decomposition: R1 = (C,A), R21 = (D, B) and R22 = (D, C)

- The decomposition does not preserve the dependencies $AB \rightarrow C$, $AB \rightarrow D$ in F

(We can verify this by calculating the F1, F21 and F22 from F+, performing the union and calculating the closure of this union). We can also verify it using the algorithm:

result = α
while (changes to result) do
for each Ri in the decomposition
t = (result ∩ Ri)+ ∩ Ri
result = result ∪ t

If *result* contains all attributes in β, then the functional dependency $\alpha \rightarrow \beta$ is preserved.

Exercise 4 - solution

$R = (A, B, C, D)$

$F = \{AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B\}$

Final decomposition: R1 = (C,A), R21 = (D, B). R22 = (D, C)

For example, for the dependency $AB \rightarrow C$:

result = AB

For CA: $t = (result \cap Ri)^+ \cap Ri = (AB \cap AC)^+ \cap AC = A^+ \cap AC = A$;

result = $result \cup t = AB$;

For DB: $t = (AB \cap BD)^+ \cap BD = B^+ \cap BD = B \cap BD = B$;

result = $AB \cup B = AB$;

For DC: $t = (AB \cap DC)^+ \cap DC = \emptyset \cap DC = \emptyset$;

result = $AB \cup \emptyset = AB$;

Exercise 5

$R = (A, B, C, D, E)$

$F = \{A \rightarrow B, BC \rightarrow E, ED \rightarrow A\}$

a. Find the candidate keys of R.
b. Is R in 3NF? If not, decompose it into 3NF.
c. Is R in BCNF? If not, decompose it into BCNF.

Exercise 5 - solution

$$R = (A, B, C, D, E)$$

$$F = \{A \rightarrow B, BC \rightarrow E, ED \rightarrow A\}$$

- We find the candidate keys. Attributes, A,B,C appear in both the LHS and the RHS of FDs. Attributes C and D appear only on the LHS of FDs. Therefore attributes C and D are certainly part of any candidate key.
- We start with the calculation of the closure of the attribute set CD.
- If we don't find a key we continue by adding one more attribute every time:
- We first calculate CD+ = CD.
- We continue by adding one more attribute in the set CD::

  - CDA+ = ABCDE – therefore CDA is a candidate key.

  - CDB+ = ABCDE – therefore CDB is a candidate key.

  - CDE+ = ABCDE – therefore CDE is a candidate key.

Exercise 5 - solution

$$R = (A, B, C, D, E)$$

$$F = \{A \rightarrow B, BC \rightarrow E, ED \rightarrow A\}$$

- Therefore the candidate keys of R are: CDA, CDB, CDE.

All FDs in F satisfy the definition of 3NF:

- in $A \rightarrow B$, the RHS is a prime attribute
- in $BC \rightarrow E$, the RHS is a prime attribute
- in $ED \rightarrow A$, the RHS is a prime attribute

Therefore R is in 3NF.

Exercise 5 - solution

$$R = (A, B, C, D, E)$$

$$F = \{A \rightarrow B, BC \rightarrow E, ED \rightarrow A\}$$

- Therefore the candidate keys of R are: CDA, CDB, CDE.

No FDs in F satisfies the definition of BCNF:

- in $A \rightarrow B$, the LHS is not a key
- in $BC \rightarrow E$, the LHS is not a key
- in $ED \rightarrow A$, the LHS is not a key

Therefore R is not in BCNF.

Exercise 5 - solution

$$R = (A, B, C, D, E)$$

$$F = \{A \rightarrow B, BC \rightarrow E, ED \rightarrow A\}$$

We decompose R in BCNF:

- We choose $BC \rightarrow E$ in order to decompose into:

  R1 = (B,C,E) and R2 = (A, B, C, D).

  R1 is in BCNF with key BC.
- We need to find the candidate keys of R2 in order to see if it is in BCNF. A+ = AB, C+ = C, B+ = B, and D+ = D (no key from the closure of singleton attributes). We check the closure of pairs of attributes: AB+ = AB, AC+ = ABC, AD+ = ABD, BC+= BC, BD+ = BD and CD+ = CD (no key from the closure of pairs of attributes). We continue with triples of attributes: ABC+ = ABC, ABD+ = ABD, BCD+ = ABCD, CDA+ = ABCD.
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Exercise 5 - solution

$R = (A, B, C, D, E)$

$F = \{A \rightarrow B, BC \rightarrow E, ED \rightarrow A\}$

Decomposition until this point: R1 = (B,C,E) and R2 = (A, B, C, D).

- Therefore the keys of R2 are CDA and CDB. (Note that we expected this result from our knowledge about the keys of R itself.)
- We need to find F+ and the subset of F+, that holds in R2 i.e. F2+.

$F = \{A \rightarrow B, BC \rightarrow E, ED \rightarrow A\}$

$F+ = \{A \rightarrow B, BC \rightarrow E, ED \rightarrow A, ED \rightarrow B, BCD \rightarrow ED, BCD \rightarrow A, AC \rightarrow BC, AC \rightarrow E \text{ etc}\}$

$F2+ = \{A \rightarrow B, BCD \rightarrow A \text{ etc}\}$. (Note that we could find that the candidate keys of R2 are indeed CDA and CDB by observing F2+ and concluding that D and C are certainly part of any candidate key and finding the closures of CD+, CDA+ and CDB+).

Exercise 5 - solution

$$R = (A, B, C, D, E)$$

$$F = \{A \rightarrow B, BC \rightarrow E, ED \rightarrow A\}$$

Decomposition until this point: R1 = (B,C,E) and R2 = (A, B, C, D).

The keys of R2 are CDA and CDB.

- We found F2+ based on the calculation of F+. Note that, alternatively we could use the algorithm:

To check if a relation R_i in a decomposition of R is in BCNF:

1. Use the original set of dependencies F that hold on R for every set of attributes $\alpha \subseteq$ Ri, check that α+ either includes no attribute of Ri-α, or includes all attributes of Ri
2. If the condition is violated by some $\alpha \rightarrow \beta$ in F, the dependency

   $\alpha \rightarrow (\alpha^+ - \alpha) \cap R_i$

   can be shown to hold on Ri, and Ri violates BCNF.

Exercise 5 - solution

$$R = (A, B, C, D, E)$$
$$F = \{A \rightarrow B, BC \rightarrow E, ED \rightarrow A\}$$

Decomposition until this point: R1 = (B,C,E) and R2 = (A, B, C, D).

The keys of R2 are CDA and CDB.

- Using this algorithm, we calculate A+ based on F: A+ = AB. Both attributes belong to R, therefore A → B holds on R2 (indeed, since it holds in R it should also hold on R2).
- (By the way, if we use the algorithm exhaustively, we will also find that B+ = B, C+ = C, D+ = D, AB+ = AB, BC+ = BCE ∩ R2 = BC, CD+ = CD, AC+=ABCE ∩ R2 = BC, AD+ = ABD, BD+ = BD, ABC+ = ABC, ABD+ = ABD, BCD+ = BCDE ∩ R2 = BCD. Therefore, the only dependency that violates BCNF is A → B, note that AC → BC, AD → BD are essentially the same dependency.)

Exercise 5 - solution

$$R = (A, B, C, D, E)$$

$$F = \{A \rightarrow B, BC \rightarrow E, ED \rightarrow A\}$$

Decomposition until this point: R1 = (B, C, E) and R2 = (A, B, C, D) and F2+ = {A→B, BCD → A etc}.

- Therefore R2 is not in BCNF because in A → B the LHS is not a key.
- We decompose R2 to: R21 = (A, B) and R22 = (A, C, D). The key of R21 is A, the key of R22 is CDA and both are in BCNF,
- The final decomposition is: R1 = (B, C, E), R21 = (A, B) and R22 = (A, C, D).

Exercise 6

$$R = (A, B, C, D, E, G)$$

$$F = \{AB \rightarrow C, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, BC \rightarrow A, E \rightarrow G\}$$

Assume the following decompositions of R:

(a) {AB, BC, ABDE, EG}

(b) {ABC, ACDE, ADG}

For the above, answer the following:

i. Is the decomposition lossless join?

ii. Is the decomposition dependency preserving?

Exercise 6 - solution

$$R = (A, B, C, D, E, G)$$

$$F = \{AB \rightarrow C, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, BC \rightarrow A, E \rightarrow G\}$$

(a) {AB, BC, ABDE, EG}

i. The decomposition is not lossless. To prove this consider the following instance of R:

{(a1, b, c1, d1, e1, g1), (a2, b, c2, d2, e2, g2)}

Based on BC → A and AB → C, a1 ≠ a2 iff c1 ≠ c2. It is easy to see that the join AB ⋈ BC contains 4 tuples:

{(a1, b, c1), (a1, b, c2), (a2, b, c1), (a2, b, c2)}

Therefore the join of AB, BC, ABDE and EG will contain at least 4 tuples. Therefore this is not a lossless join decomposition.

Exercise 6 - solution

$$R = (A, B, C, D, E, G)$$

$$F = \{AB \rightarrow C, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, BC \rightarrow A, E \rightarrow G\}$$

(a) {AB, BC, ABDE, EG}

i. We could verify this by considering the candidate keys of AB and BC and the formal guarantee of lossless decomposition that requires that "$R1 \cap R2 \rightarrow R1$ or $R1 \cap R2 \rightarrow R2$" is a necessary and sufficient condition in case that the constraints are only functional dependencies.

Based on F, B+ does not give us either AB or BC, therefore it is not the key for either AB or BC. Therefore, the decomposition is not lossless join.

Exercise 6 - solution

$R = (A, B, C, D, E, G)$

$F = \{AB \rightarrow C, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, BC \rightarrow A, E \rightarrow G\}$

(a) {AB, BC, ABDE, EG}

ii. This is not a dependency preserving decomposition, since it does not preserve the dependency $AB \rightarrow C$:

result = AB

For AB: $t = (result \cap Ri)+ \cap Ri = (AB \cap AB)+ \cap AB = ABCDEG \cap AB = AB$;

result = result $\cup$ t = AB;

For BC: $t = (AB \cap BC)+ \cap BC = B+ \cap BC = BD \cap BC = B$;

result = $AB \cup B = AB$;

For ABDE: $t = (AB \cap ABDE)+ \cap ABDE = AB+ \cap ABDE = ABD$;

result = $AB \cup ABD = ABD$;

For EG: $t = (ABD \cap EG)+ \cap ABDE = \emptyset \cap ABDE = \emptyset$;

result = $ABD \cup \emptyset = ABD$;

Exercise 6 - solution

$$R = (A, B, C, D, E, G)$$

$$F = \{AB \rightarrow C, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, BC \rightarrow A, E \rightarrow G\}$$

(a) {AB, BC, ABDE, EG}

ii. This is not a dependency preserving decomposition, since it does not preserve the dependency AB → C:

The result does not contain C, therefore the dependency is not preserved.

Note that there is another way to test dependency preservation, and that is to calculate F+, calculate the subsets of F1, F2, F3 and F4 of F+ onto AB, BC, ABDE and EG, calculate the closure (F1 ∪ F2 ∪ F3 ∪ F4)+ and find that it is not equal to F+. However, calculating this F+ is not practical by hand, as it is very large even without the trivial dependencies.

https://arjo129.github.io/functionalDependencyCalculator/index.html

Exercise 6 - solution

$R = (A, B, C, D, E, G)$

$F = \{AB \rightarrow C, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, BC \rightarrow A, E \rightarrow G\}$

(b) {ABC, ACDE, ADG}

i. The decomposition is lossless join.

The join of ABC, ACDE and ADG can be done in 2 steps:

J1 = ABC ⋈ ACDE: this is lossless because the intersection is:

ABC ∩ ACDE =AC is the key of ABC based on the dependency AC →B.

J2 = J1 ⋈ ADG is lossless because the intersection is ABCDE ∩ ADG =AD, which is the key of ADG based on the dependencies

AD →E and E →G.

Hence the decomposition is lossless.

Exercise 6 - solution

$R = (A, B, C, D, E, G)$

$F = \{AB \rightarrow C, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, BC \rightarrow A, E \rightarrow G\}$

(b) {ABC, ACDE, ADG}

ii. This is not a dependency preserving decomposition, since it does not preserve the dependency $E \rightarrow G$:

result = E

For ABC: t = (result $\cap$ Ri)+ $\cap$ Ri = E $\cap$ AB)+ $\cap$ AB = $\emptyset$ $\cap$ AB = $\emptyset$;

result = result $\cup$ t = E;

For ACDE: t = (E $\cap$ ACDE)+ $\cap$ ACDE = E+ $\cap$ ACDE = EG $\cap$ ACDE = E;

result = E $\cup$ E = E;

For ADG: t = (E $\cap$ ADG)+ $\cap$ ADG = $\emptyset$ $\cap$ ADG = $\emptyset$;

result = E $\cup$ $\emptyset$ = E;

The result does not contain G, therefore the dependency is not preserved.

Exercise 7

From 'Database Systems Concepts' practice exercises
-- the reason this exercise is included is to show a solution of dependency preservation testing using the closures of FDs—

$R = (A, B, C, D, E,)$

$F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$

Suppose that we decompose R into:
R1 = (A, B, C) and R2 = (A, D, E)

a. Show that this decomposition is a lossless-join decomposition.
b. Show that the decomposition is not a dependency-preserving.

From 'Database Systems Concepts' practice exercises

Exercise 7 - solution

$R = (A, B, C, D, E,)$

$F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$

R1 = (A, B, C) and R2 = (A, D, E).

a. Show that this decomposition is a lossless-join decomposition.

A decomposition {R1, R2} is a lossless-join decomposition if
$R1 \cap R2 \rightarrow R1$ or $R1 \cap R2 \rightarrow R2$.
For R1 = (A,B,C), R2 = (A,D,E), $R1 \cap R2 = A$.
Since A is a candidate key for R1, therefore $R1 \cap R2 \rightarrow R1$.
Therefore the decomposition is lossless.

From 'Database Systems Concepts' practice exercises

Exercise 7 - solution

$R = (A, B, C, D, E,)$

$F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$

R1 = (A, B, C) and R2 = (A, D, E).

b. Show that the decomposition is not a dependency-preserving.

The dependency $B \rightarrow D$ is not preserved.

F1, the restriction of F to R1 is: F1 = {$A \rightarrow ABC$, $A \rightarrow AB$, $A \rightarrow AC$, $A \rightarrow BC$, $A \rightarrow B$, $A \rightarrow C$, $A \rightarrow A$, $B \rightarrow B$, $C \rightarrow C$, $AB \rightarrow AC$, $AB \rightarrow ABC$, $AB \rightarrow BC$, $AB \rightarrow AB$, $AB \rightarrow A$, $AB \rightarrow B$, $AB \rightarrow C$, AC (same as AB), BC (same as AB), ABC (same as AB)}.

F2, the restriction of F to (C,D,E) is F2 = { $A \rightarrow ADE$, $A \rightarrow AD$, $A \rightarrow AE$, $A \rightarrow DE$, $A \rightarrow A$, $A \rightarrow D$, $A \rightarrow E$, $D \rightarrow D$, E (same as A), AD, AE, DE, ADE (same as A).

From 'Database Systems Concepts' practice exercises

Exercise 7 - solution

$R = (A, B, C, D, E,)$

$F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$

R1 = (A, B, C) and R2 = (A, D, E).

b. Show that the decomposition is not dependency-preserving.

(F1 ∪ F2)+ is easily seen not to contain B → D since the only FD in F1 ∪ F2 with B as the LHS is B → B, a trivial FD. Thus B → D is not preserved. Note that CD → ABCDE is also not preserved.

A simpler argument is as follows: F1 contains no dependencies with D on the RHS. F2 contains no dependencies with B on the LHS. Therefore for B → D to be preserved there must be an FD B → α in F1+ and α → D in F2+ (so B → D would follow by transitivity). Since the intersection of the two schemes is A, α = A. Observe that B → A is not in F1+ since B+ =BD.